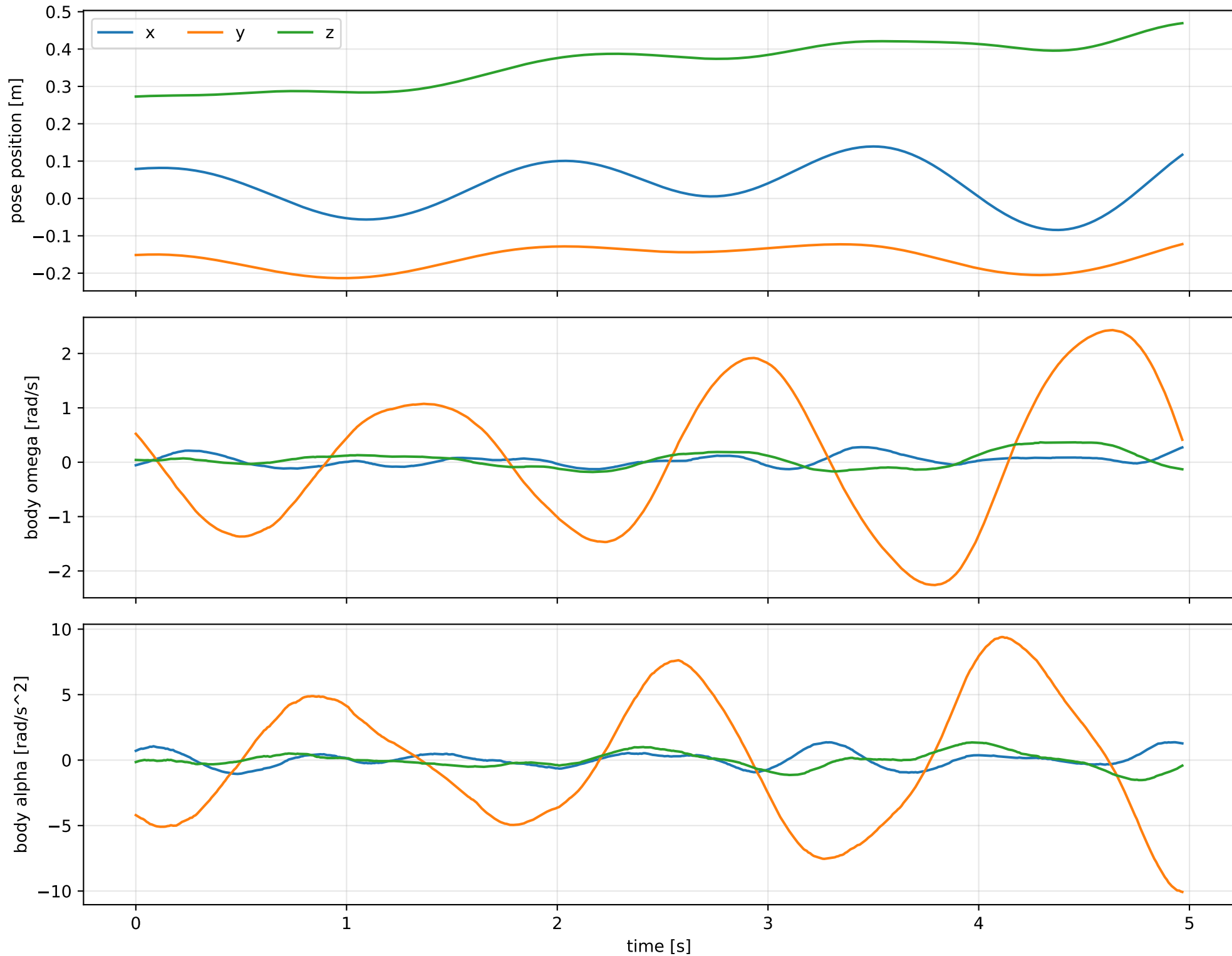
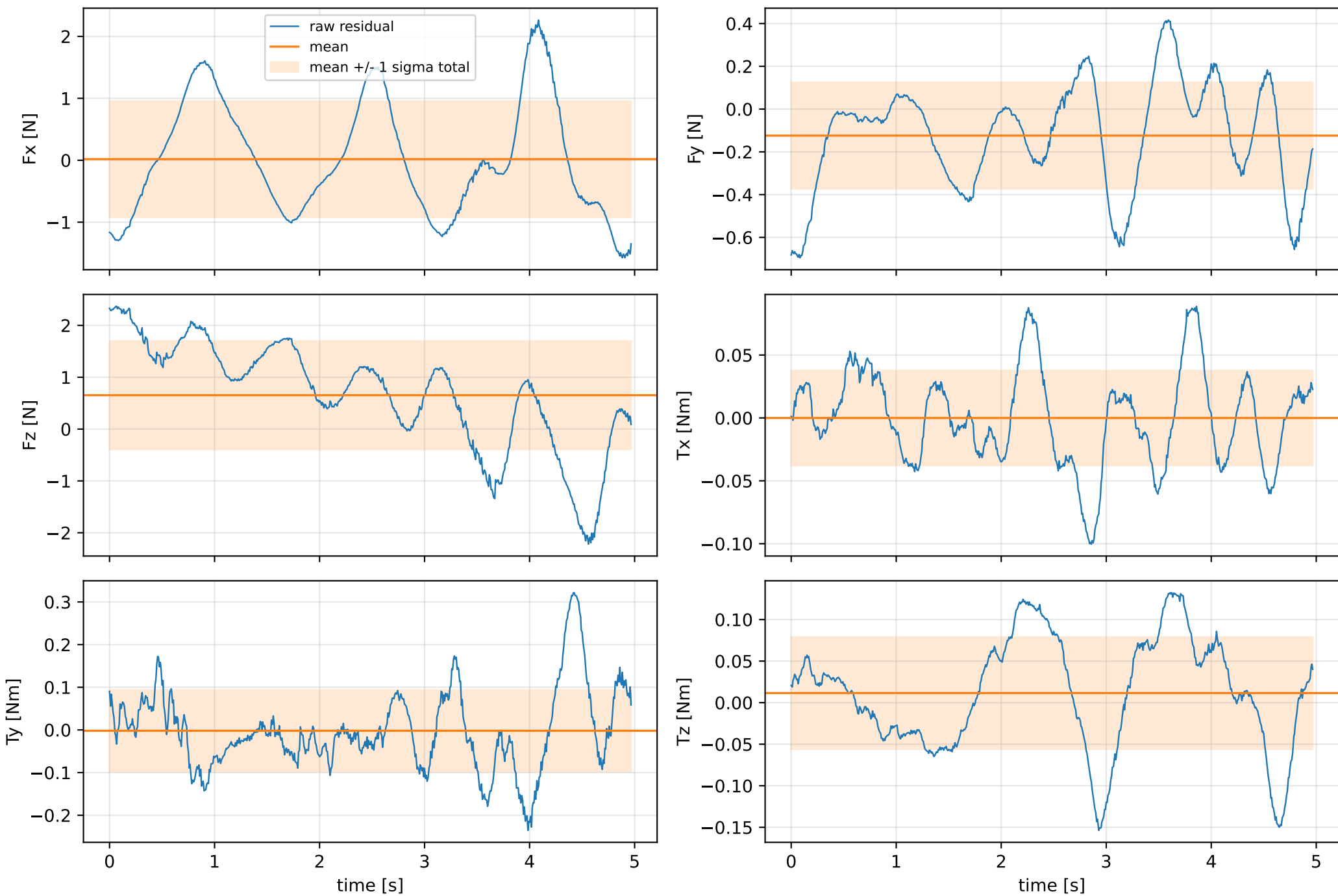


inertia_over_mass_all_nominal: SG trajectory and derived motion



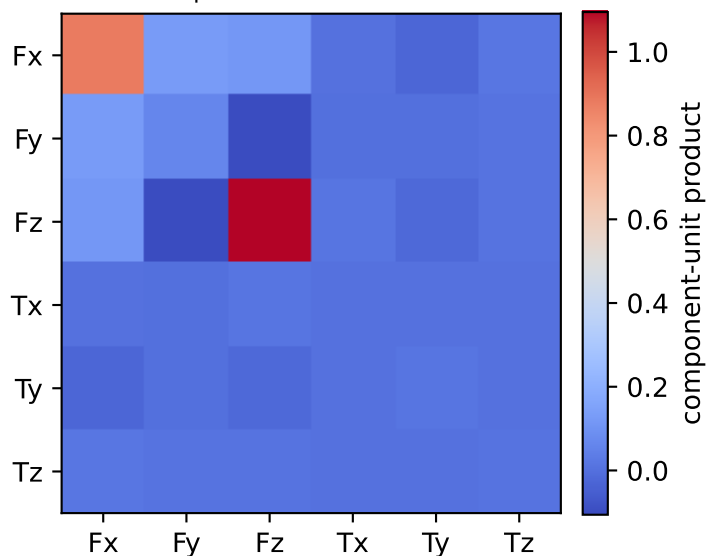
inertia_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



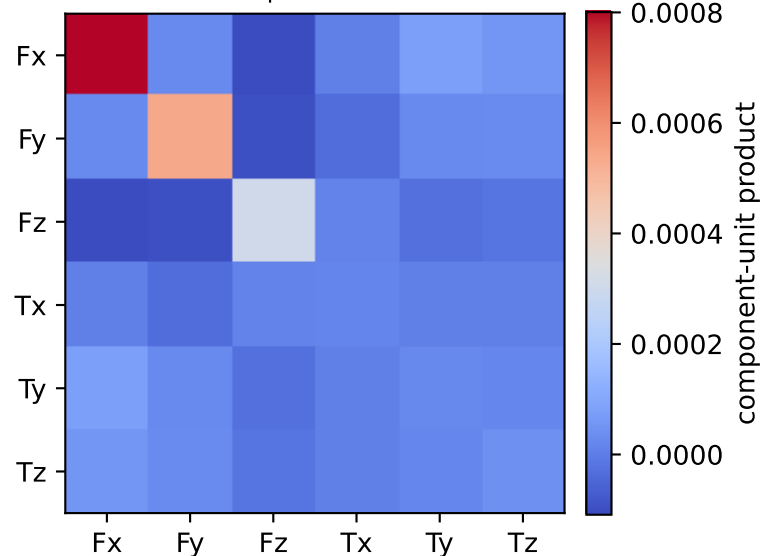
vector RMS about zero: force 1.5726 N, torque 0.12318 Nm; component std about mean: force [0.9364 0.249 1.0465], torque [0.0377 0.0953 0.0675]

inertia_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [8.456e-04 4.517e-03 6.821e-03 2.915e-02 8.608e-01 1.145e+00]

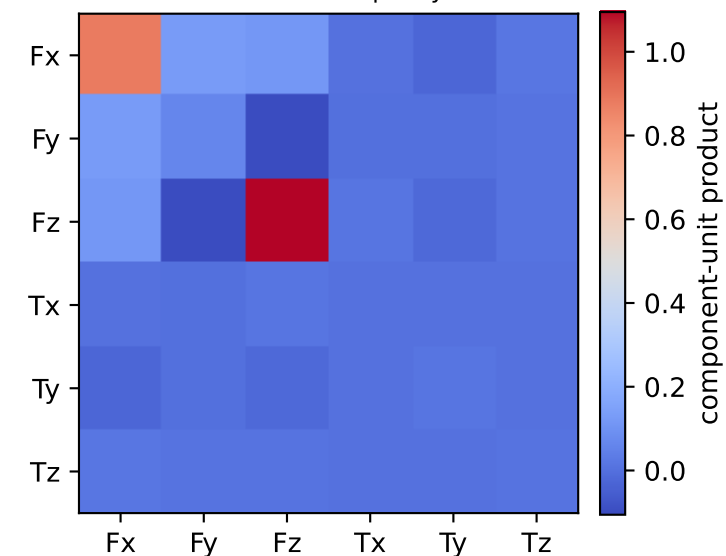
empirical total covariance



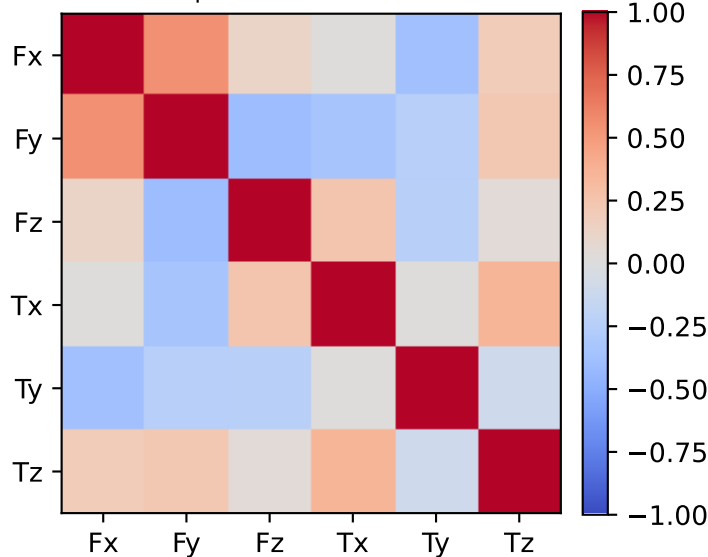
mean full-SG prediction covariance



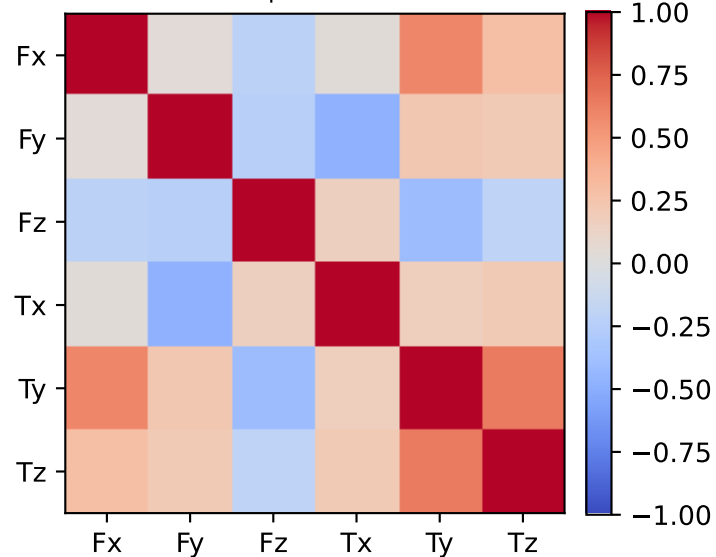
PSD excess model discrepancy covariance



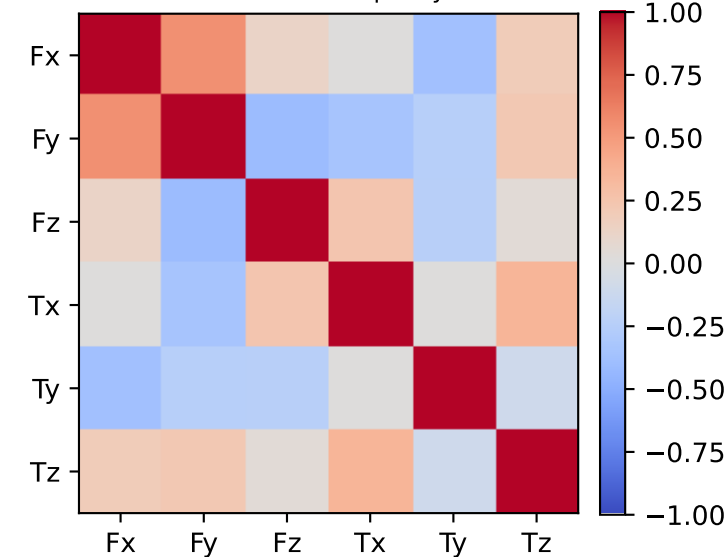
empirical total correlation



mean full-SG prediction correlation

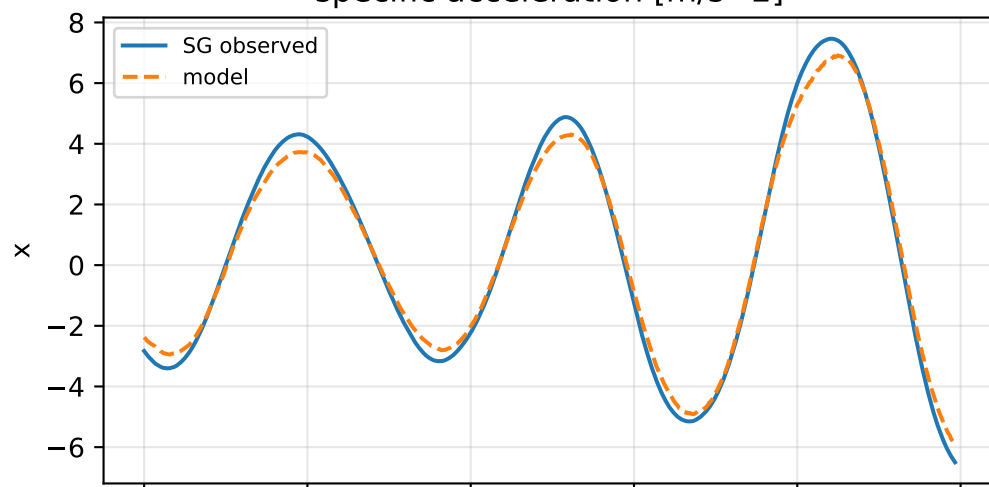


PSD excess model discrepancy correlation

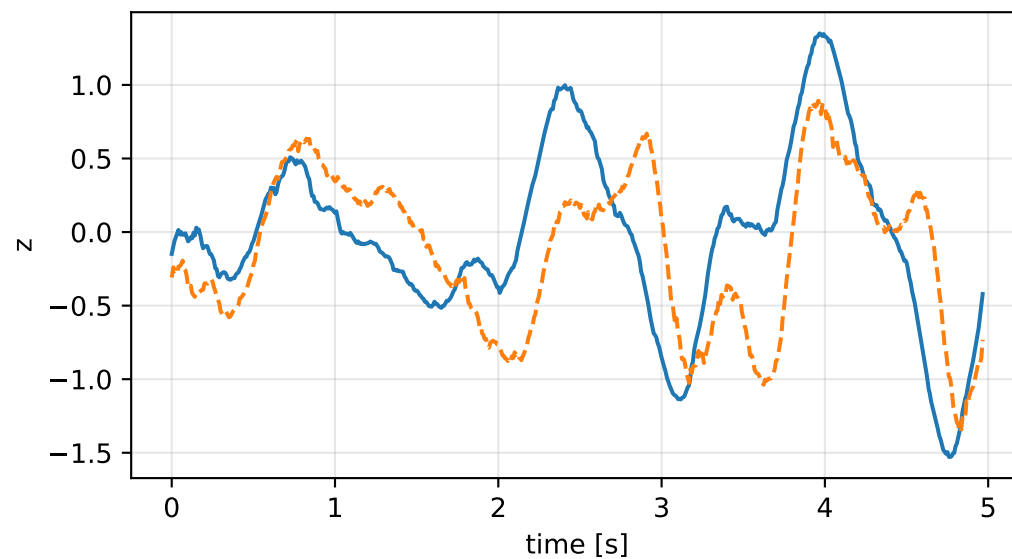
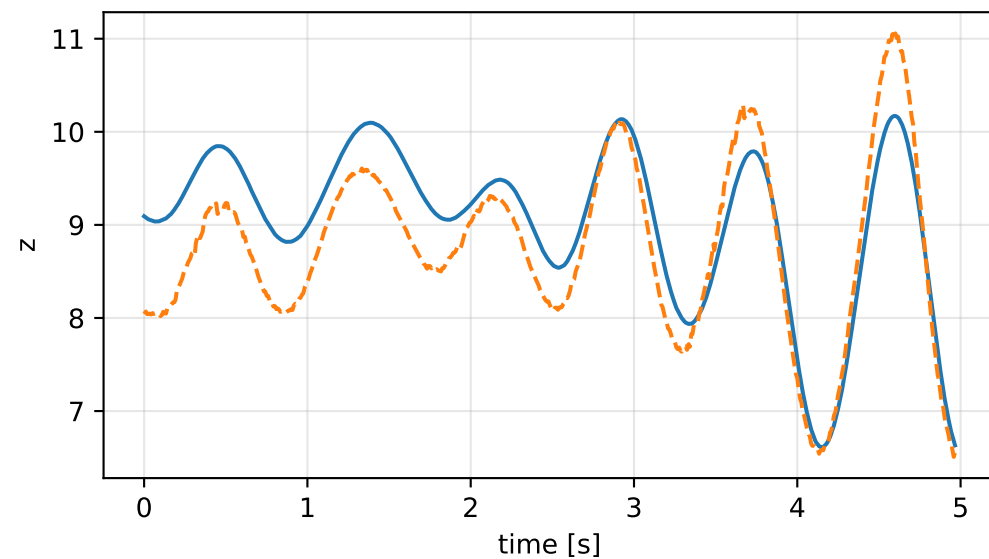
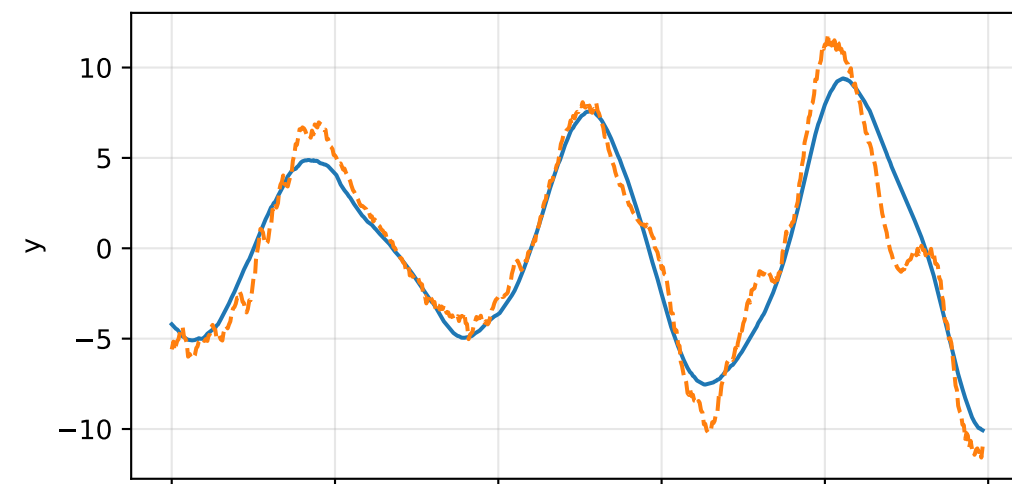
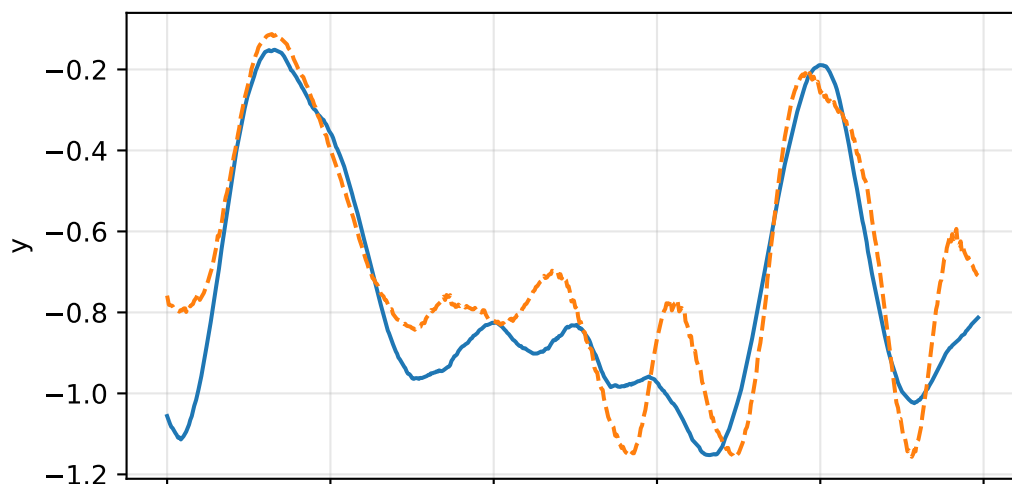
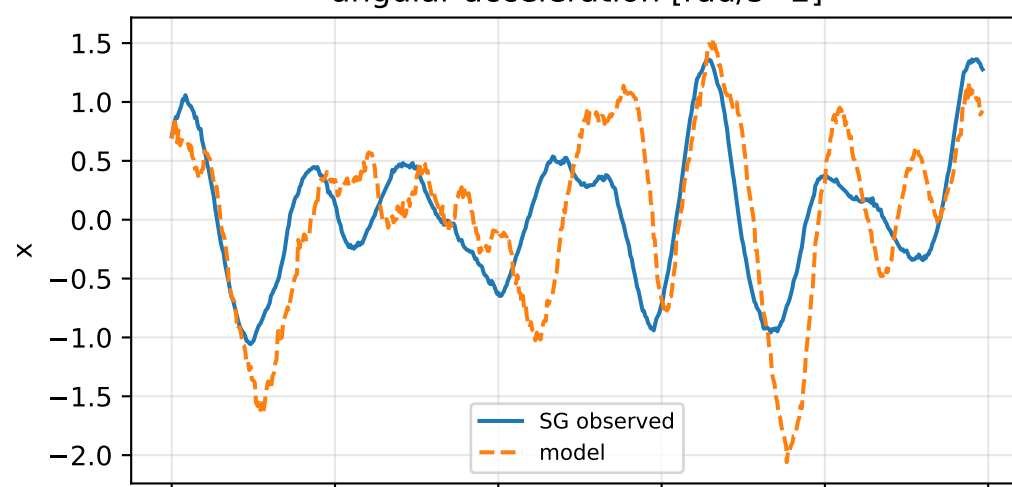


inertia_over_mass_all_nominal: acceleration residual objective

specific acceleration [m/s²]

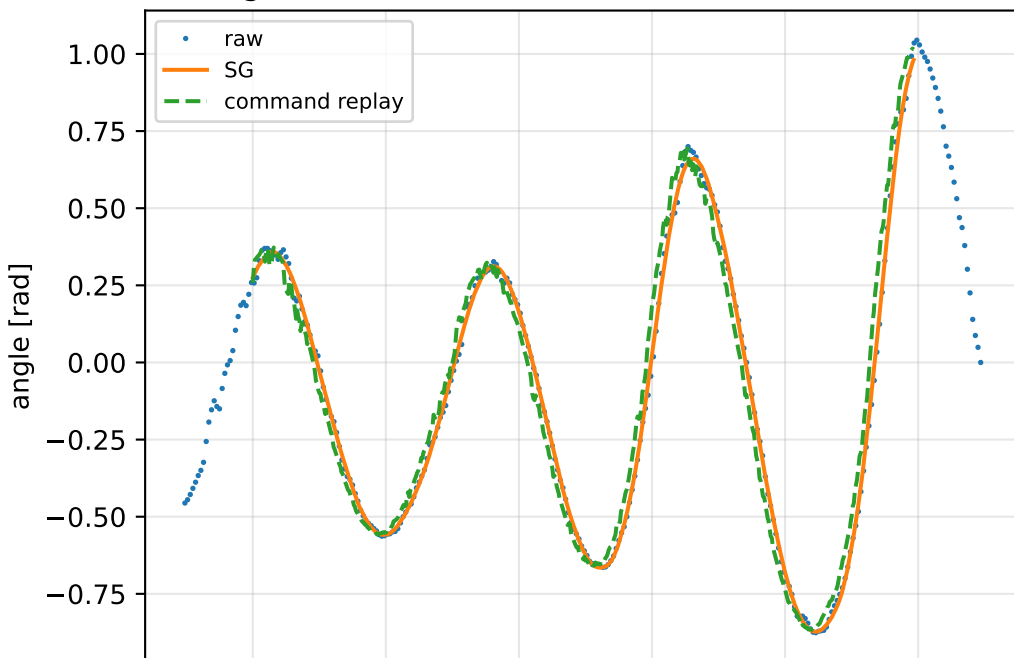


angular acceleration [rad/s²]

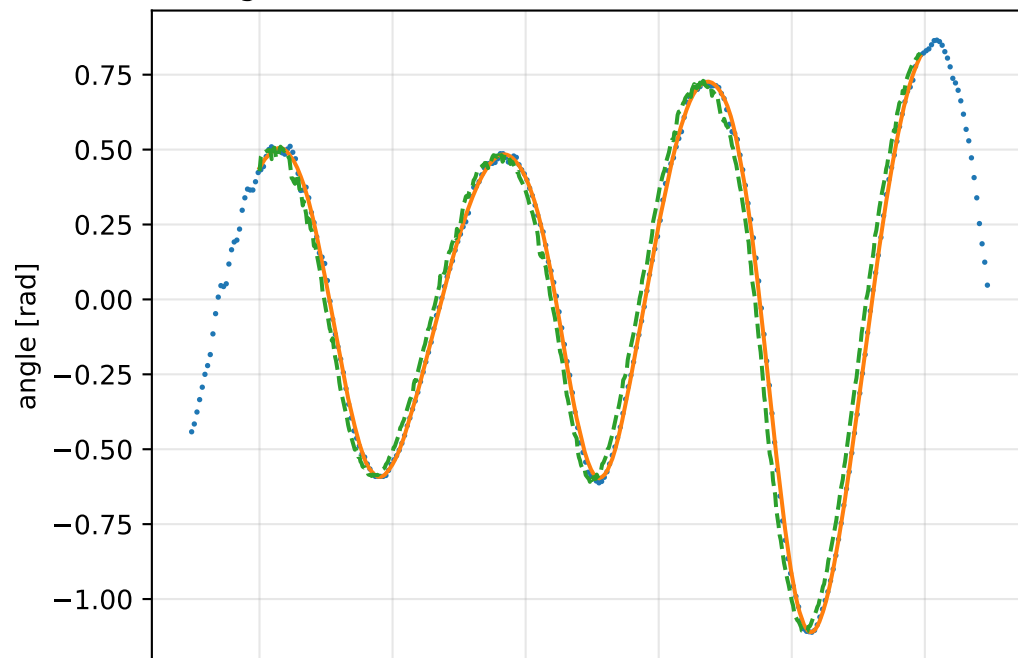


inertia_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)

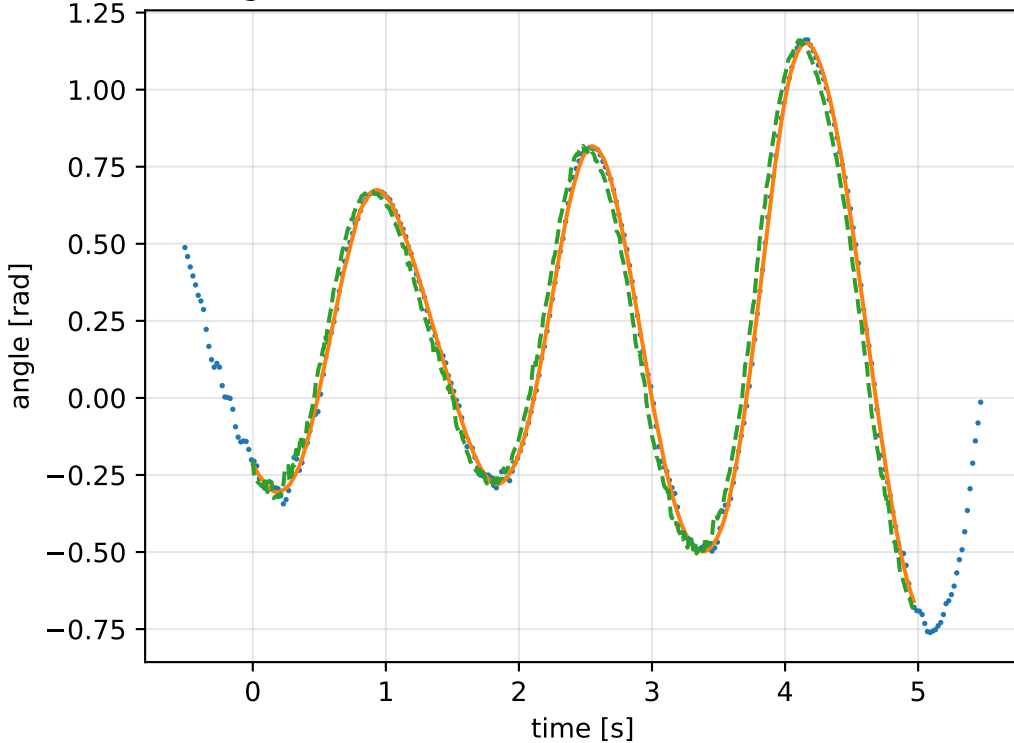
gimbal 1: RMSE=0.0759, max=0.2095 rad



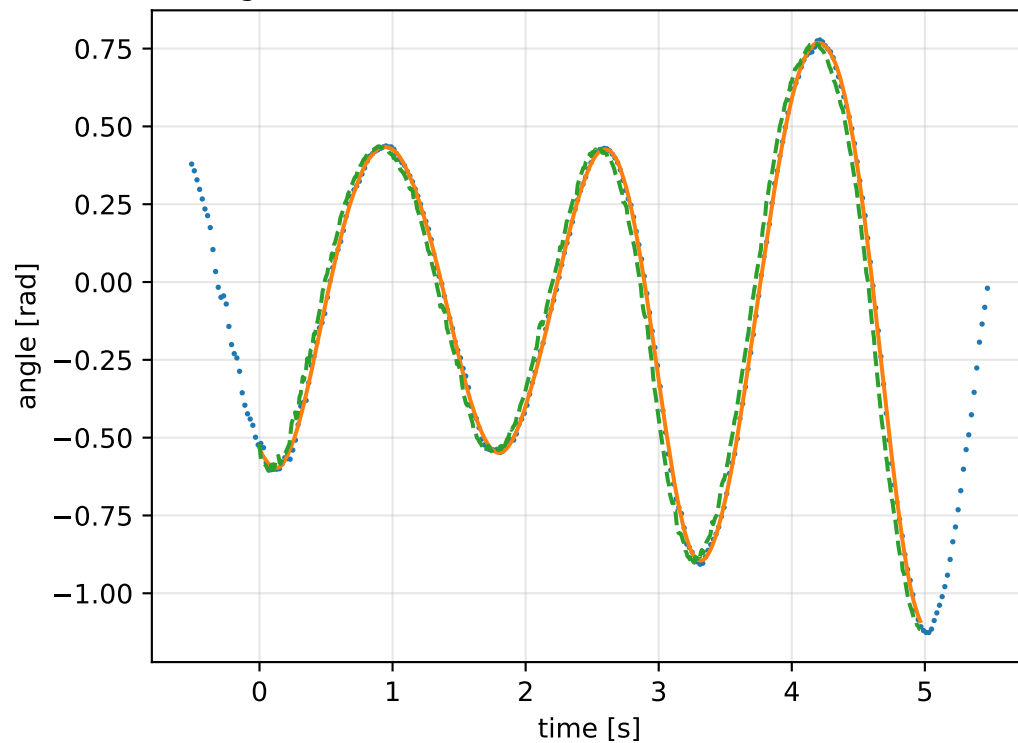
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

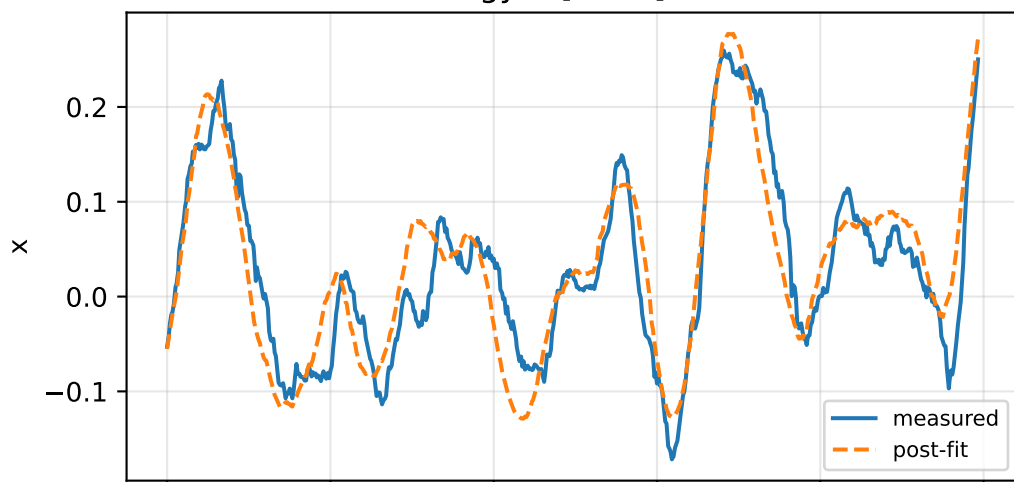


gimbal 4: RMSE=0.07131, max=0.1737 rad

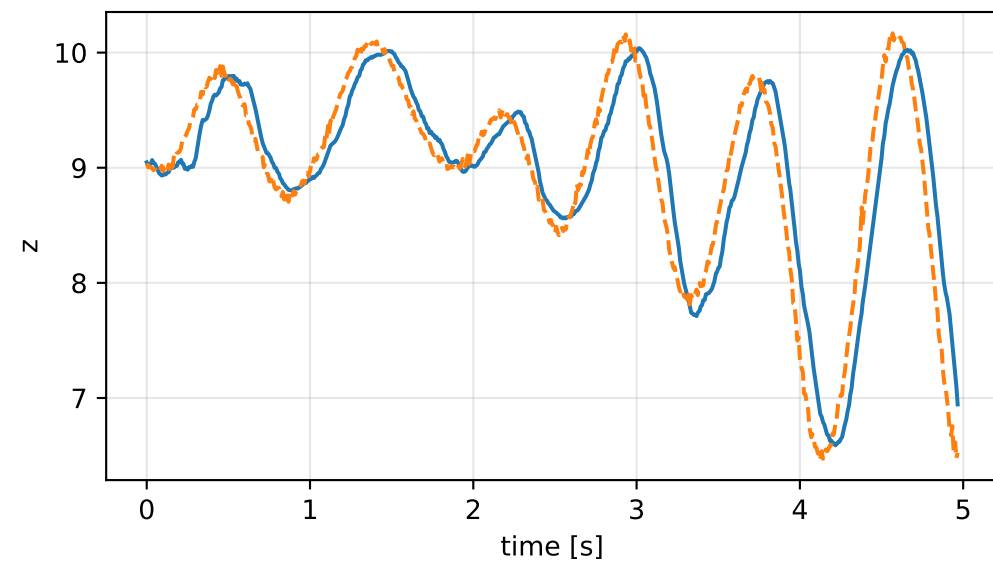
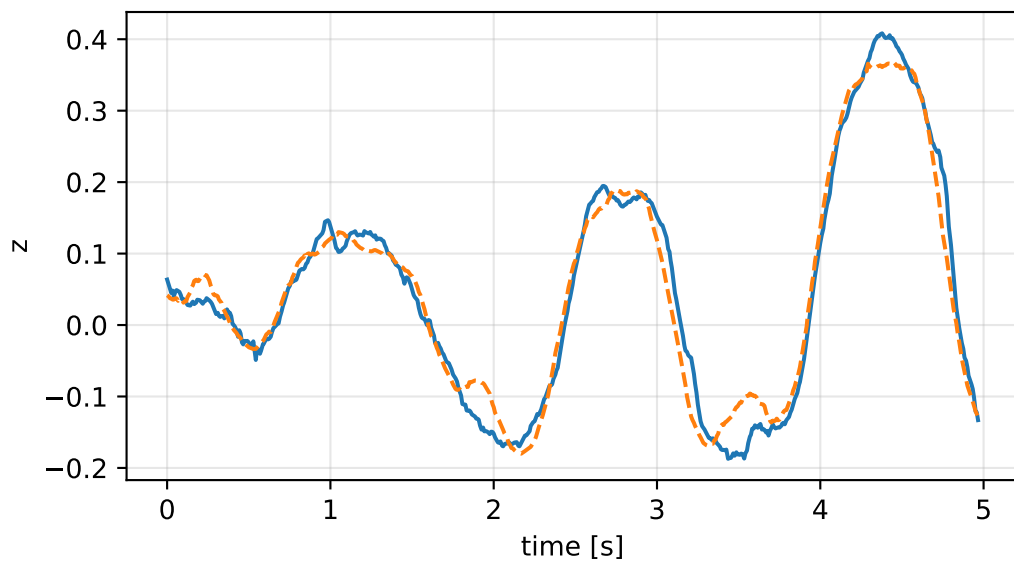
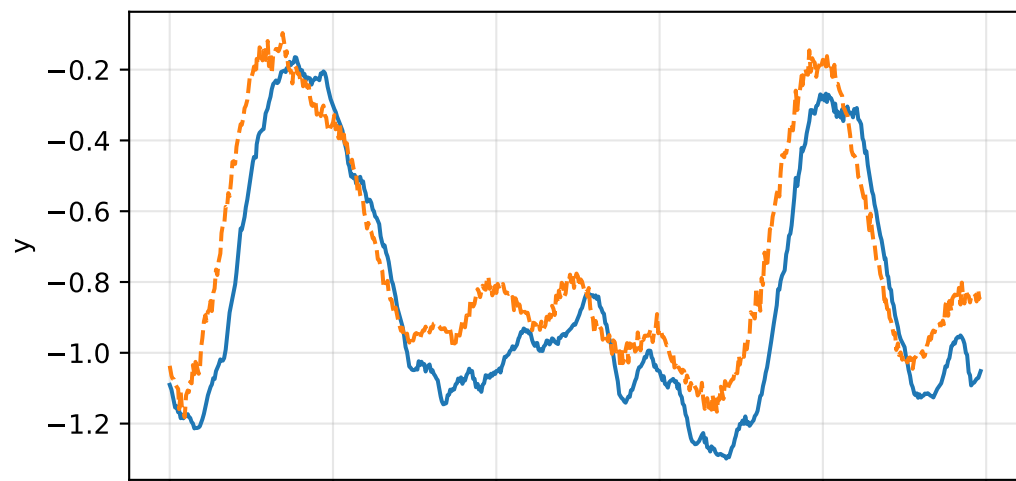
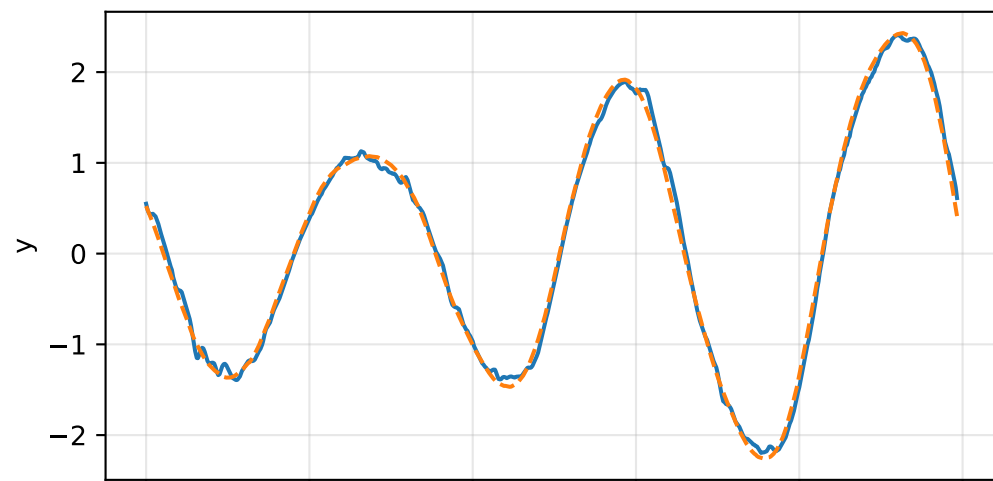
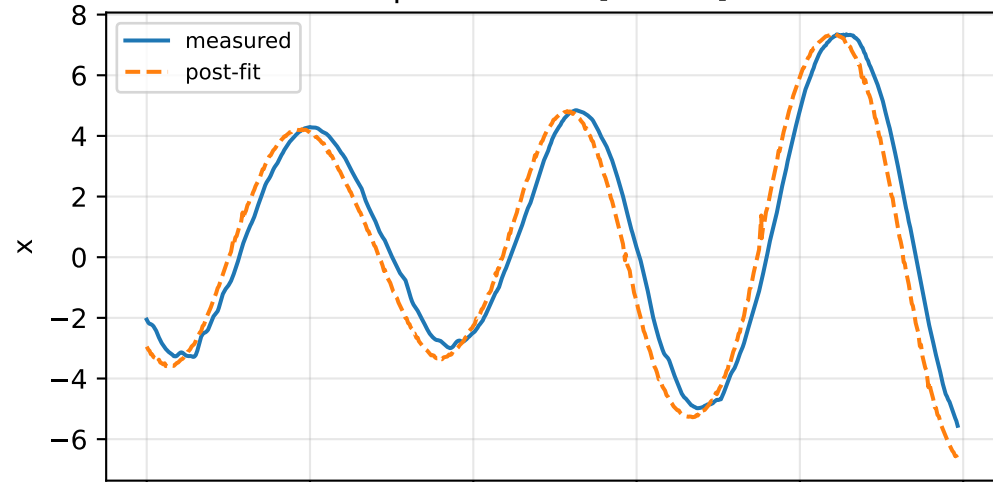


inertia_over_mass_all_nominal: IMU comparison (diagnostic only)

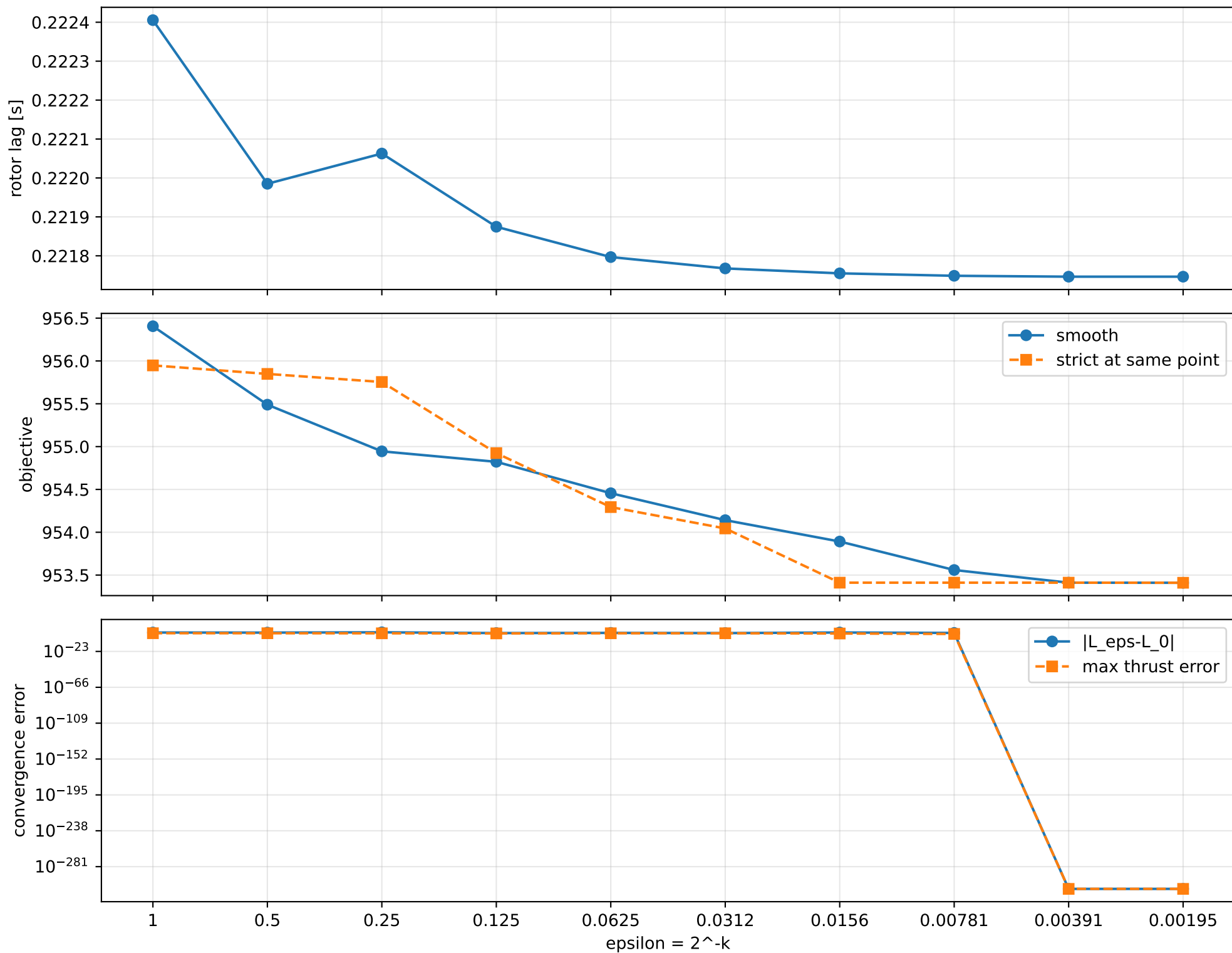
gyro [rad/s]



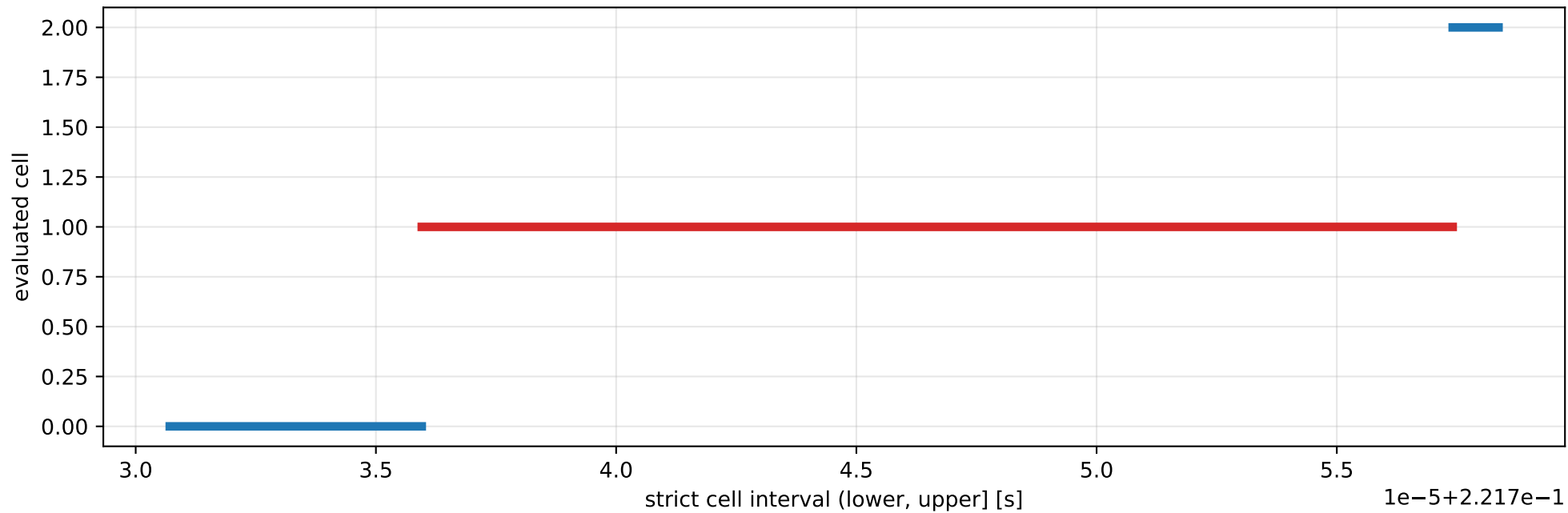
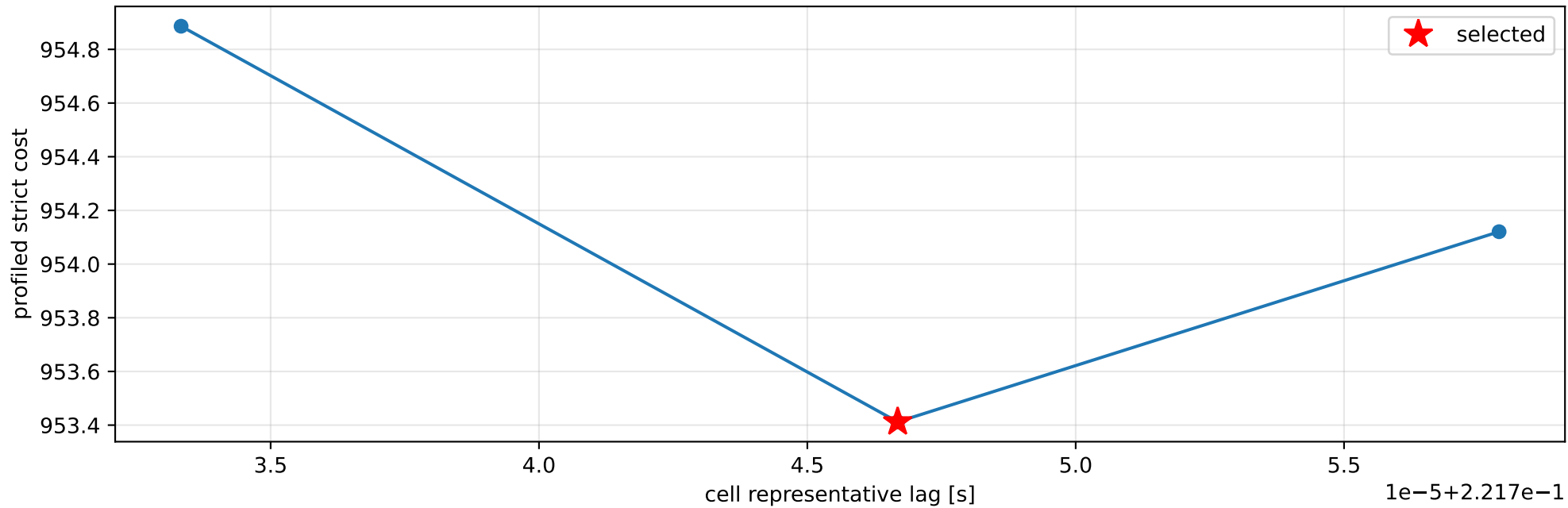
specific force [m/s²]



inertia_over_mass_all_nominal: smooth-to-strict continuation



inertia_over_mass_all_nominal: exact strict-ZOH lag cells



selected (0.221735954, 0.221757412] s; final smooth 0.221746393 s; data boundary=False

inertia_over_mass_all_nominal: parameter estimate

Case: inertia_over_mass_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.60099097

inertia [kg m²]:

[[7.18935113e-02 -8.17151651e-07 2.10332789e-05]

[-8.17151651e-07 7.18412263e-02 7.34709250e-08]

[2.10332789e-05 7.34709250e-08 1.42672083e-01]]

CoG body [m]: [2.45687205e-02 7.70487839e-05 2.38570533e-02]

force effectiveness: [0.7967463 1.05449457 0.99541175 0.79883831]

scale-free J/m [m²]:

[[2.76408155e-02 -3.14169354e-07 8.08664052e-06]

[-3.14169354e-07 2.76207135e-02 2.82472819e-08]

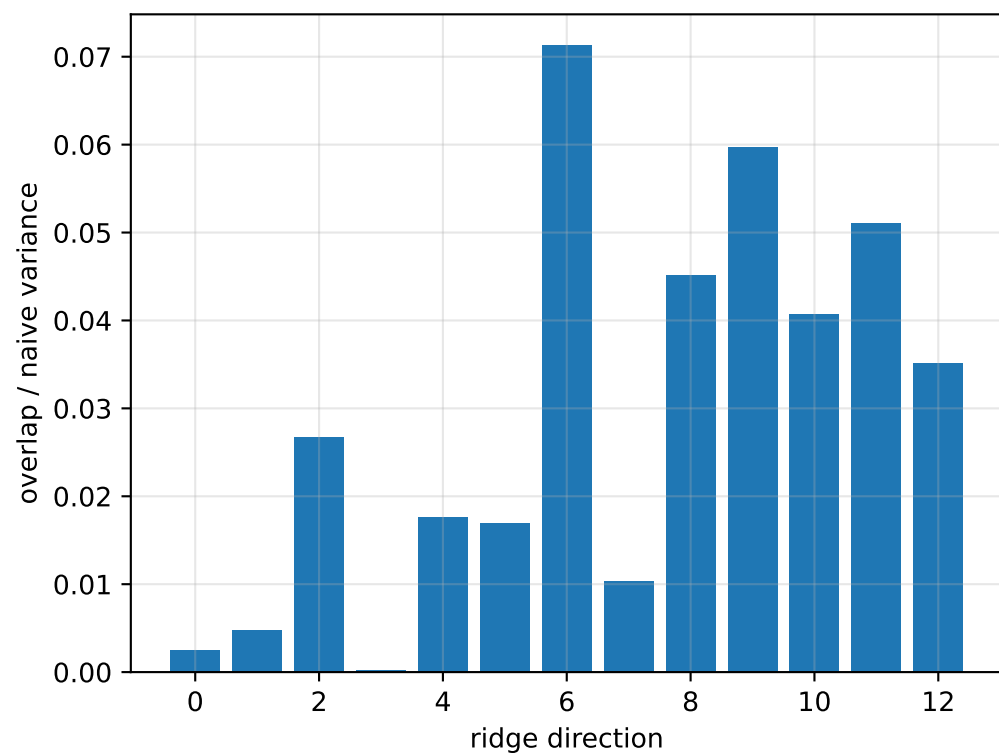
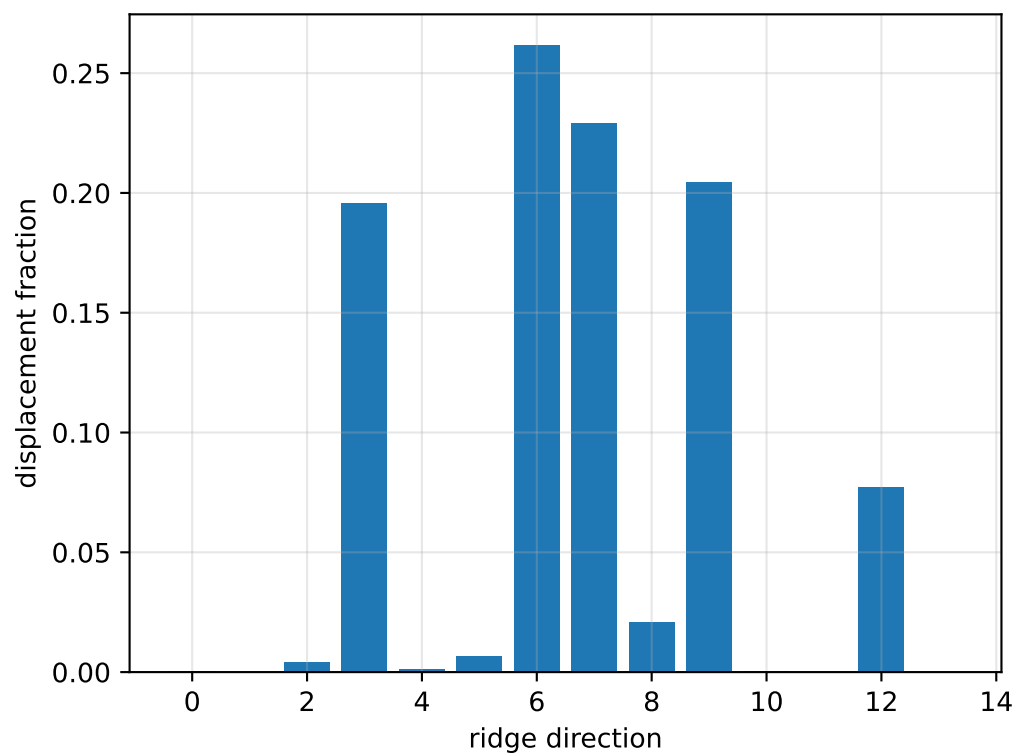
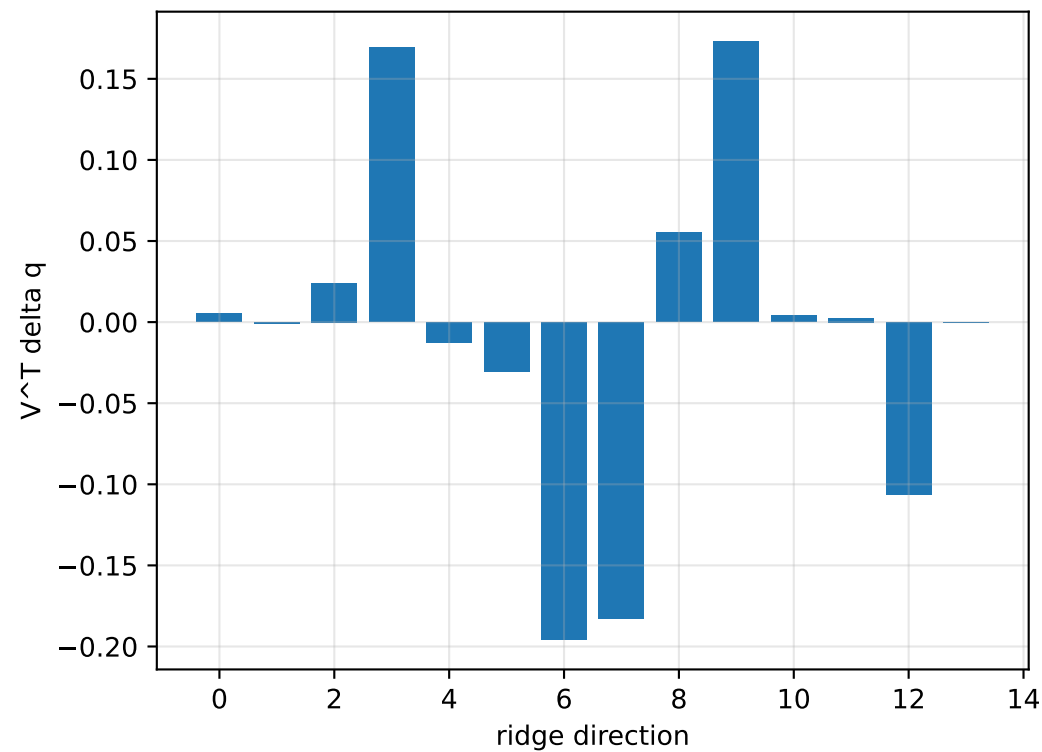
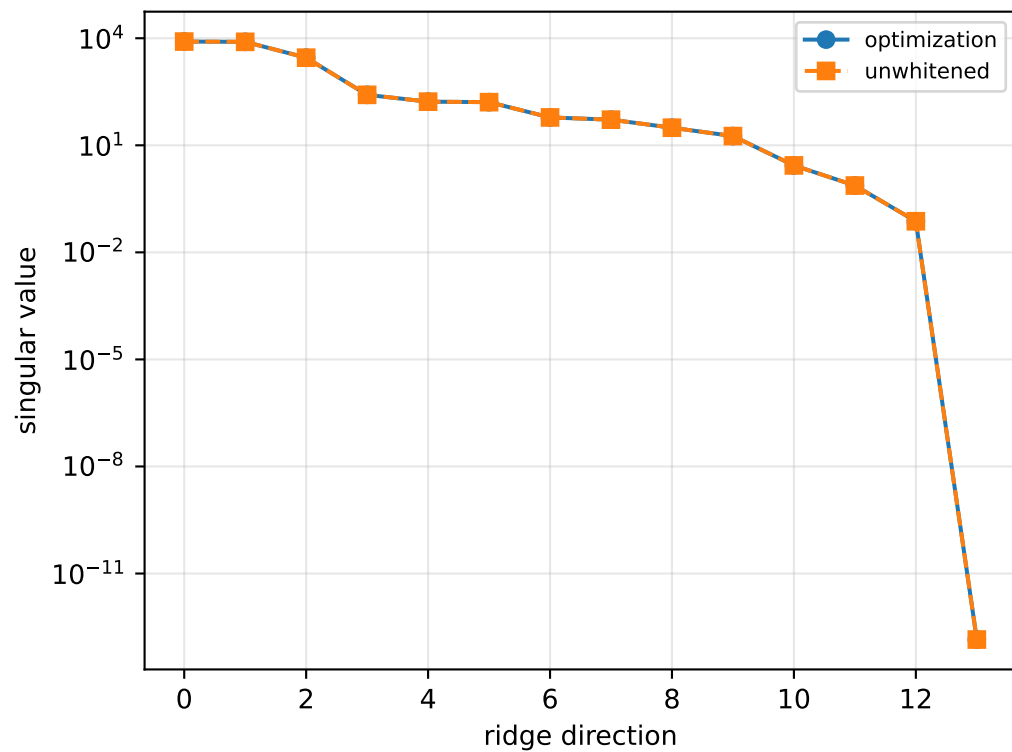
[8.08664052e-06 2.82472819e-08 5.48529713e-02]]

scale-free f/m: [0.30632413 0.40542031 0.38270481 0.30712844]

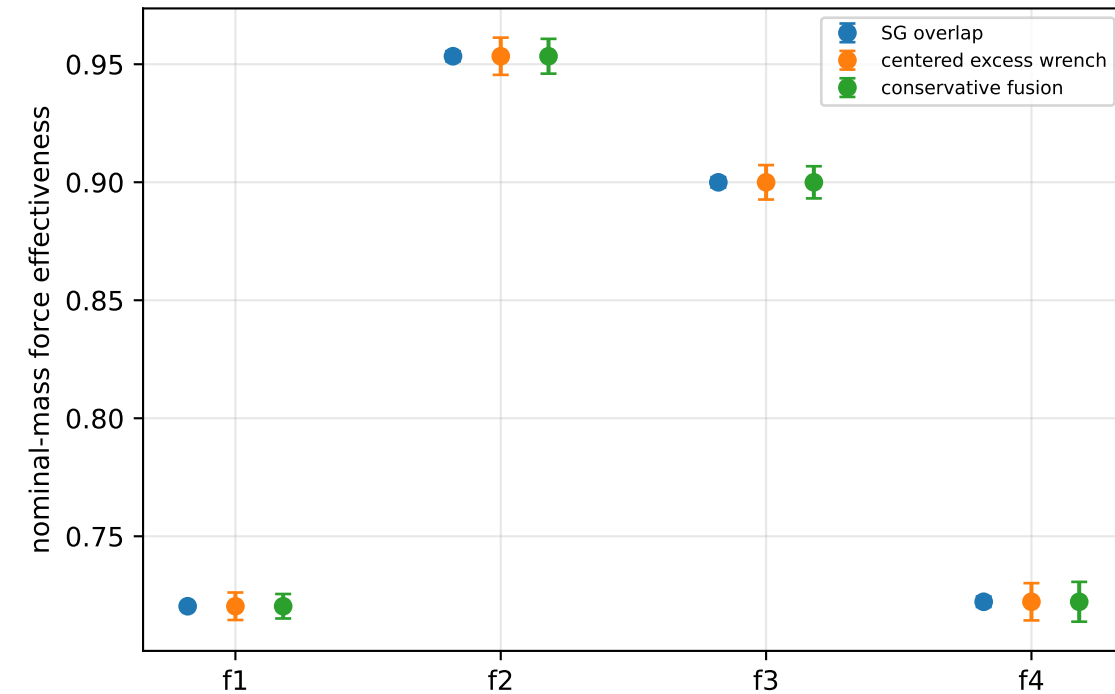
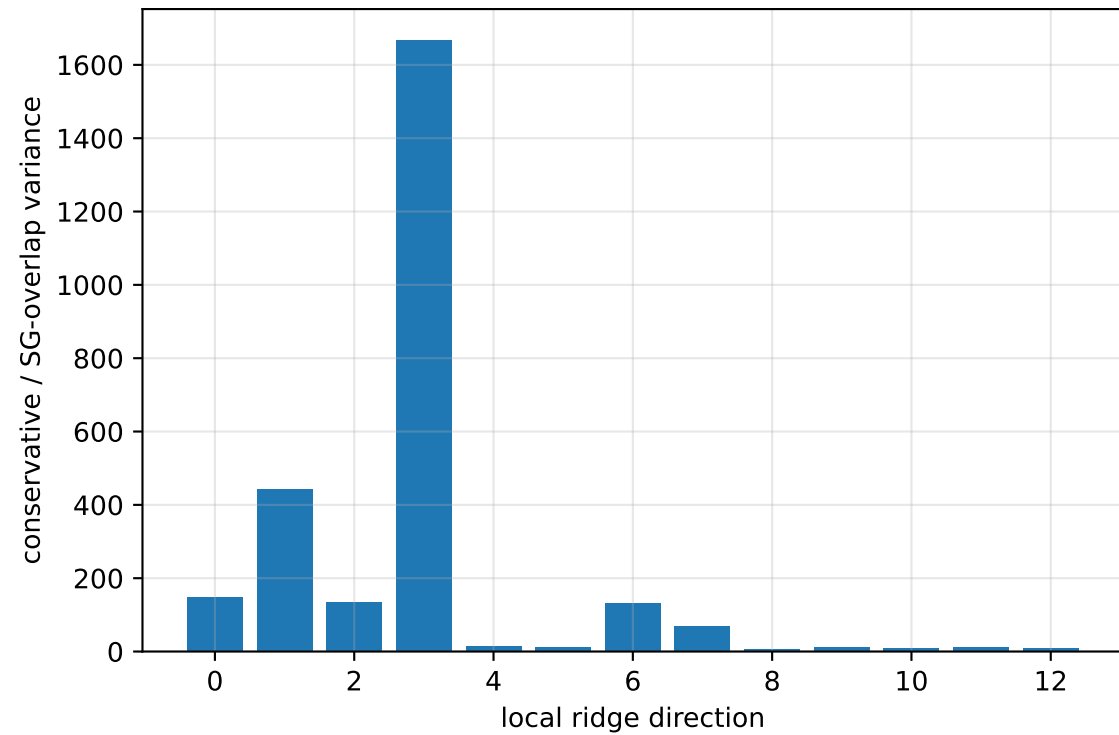
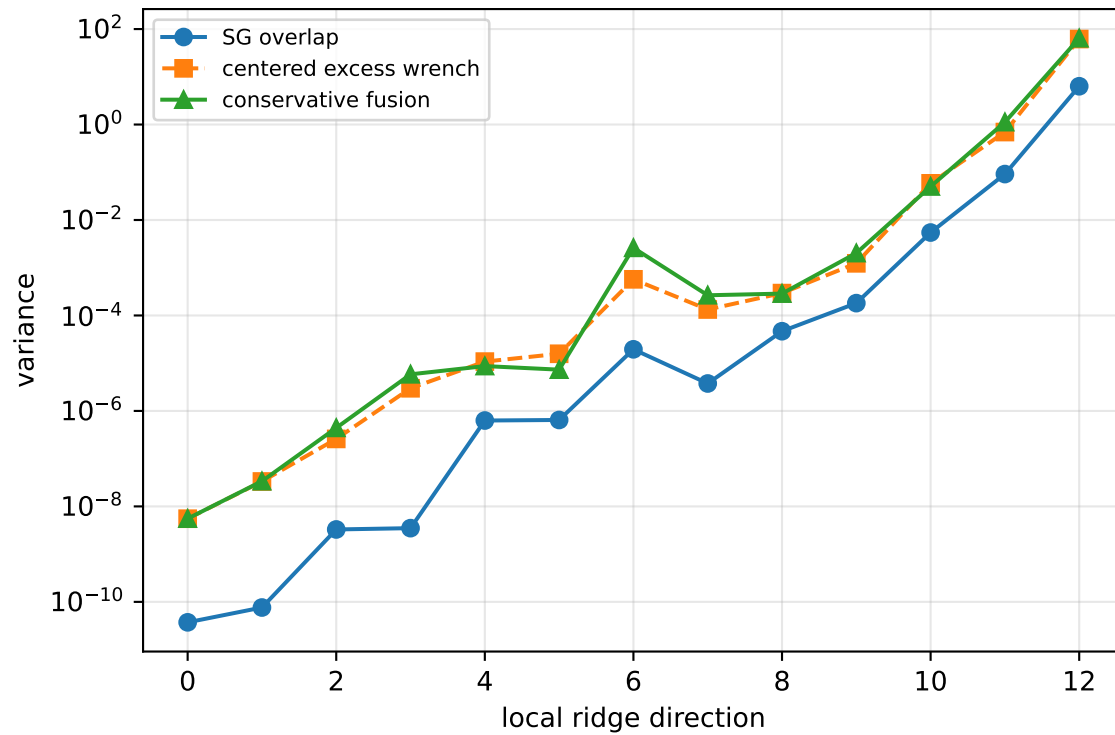
rotor lag cell: (0.221735954, 0.221757412] s

representative: 0.221746683 s

inertia_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, || $\mathbf{J}\mathbf{v}$ ||=2.64e-13)



inertia_over_mass_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 273.6$
wrench-to-acceleration recovery max error = $2.89\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

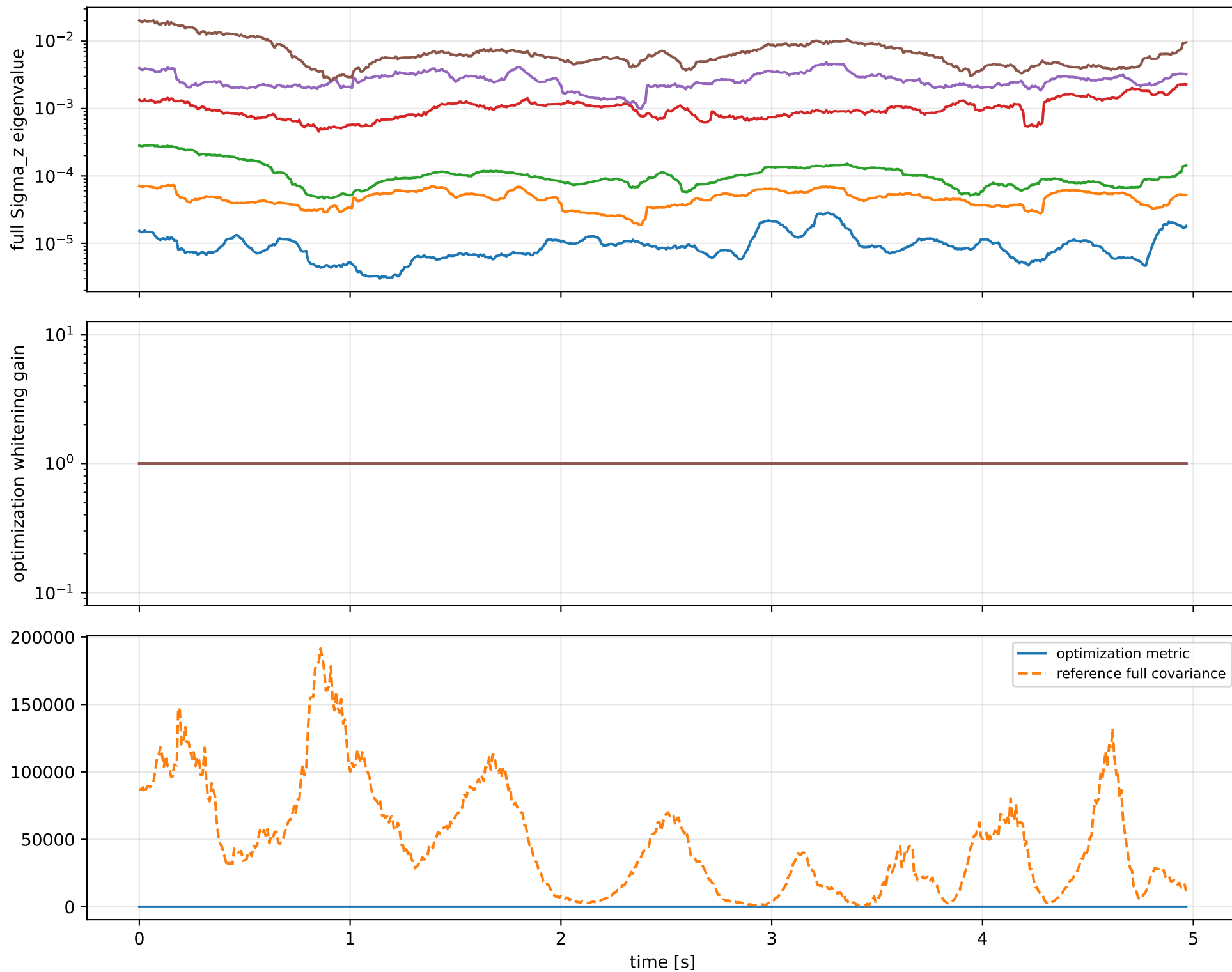
index 12: variance=64.14, ratio=10.12
 $\log_{\text{mass}}=+0.132$; second_moment_diag_1=+0.139
second_moment_diag_2=+0.138; second_moment_diag_3=-0.935
second_moment_offdiag_12=-0.000552; second_moment_offdiag_13=+0.0158
second_moment_offdiag_23=+0.0208; cog_x=+4.97e-06
cog_y=-4.17e-06; cog_z=+3.88e-06
log_force_eff_1=+0.131; log_force_eff_2=+0.132
log_force_eff_3=+0.131; log_force_eff_4=+0.132

index 11: variance=1.122, ratio=12.24
 $\log_{\text{mass}}=-0.00321$; second_moment_diag_1=-0.00267
second_moment_diag_2=+0.00397; second_moment_diag_3=+0.0147
second_moment_offdiag_12=-0.00909; second_moment_offdiag_13=+1
second_moment_offdiag_23=-0.00757; cog_x=-5.86e-05
cog_y=-0.000163; cog_z=-2.21e-06
log_force_eff_1=-0.00288; log_force_eff_2=-0.00239
log_force_eff_3=-0.00451; log_force_eff_4=-0.00299

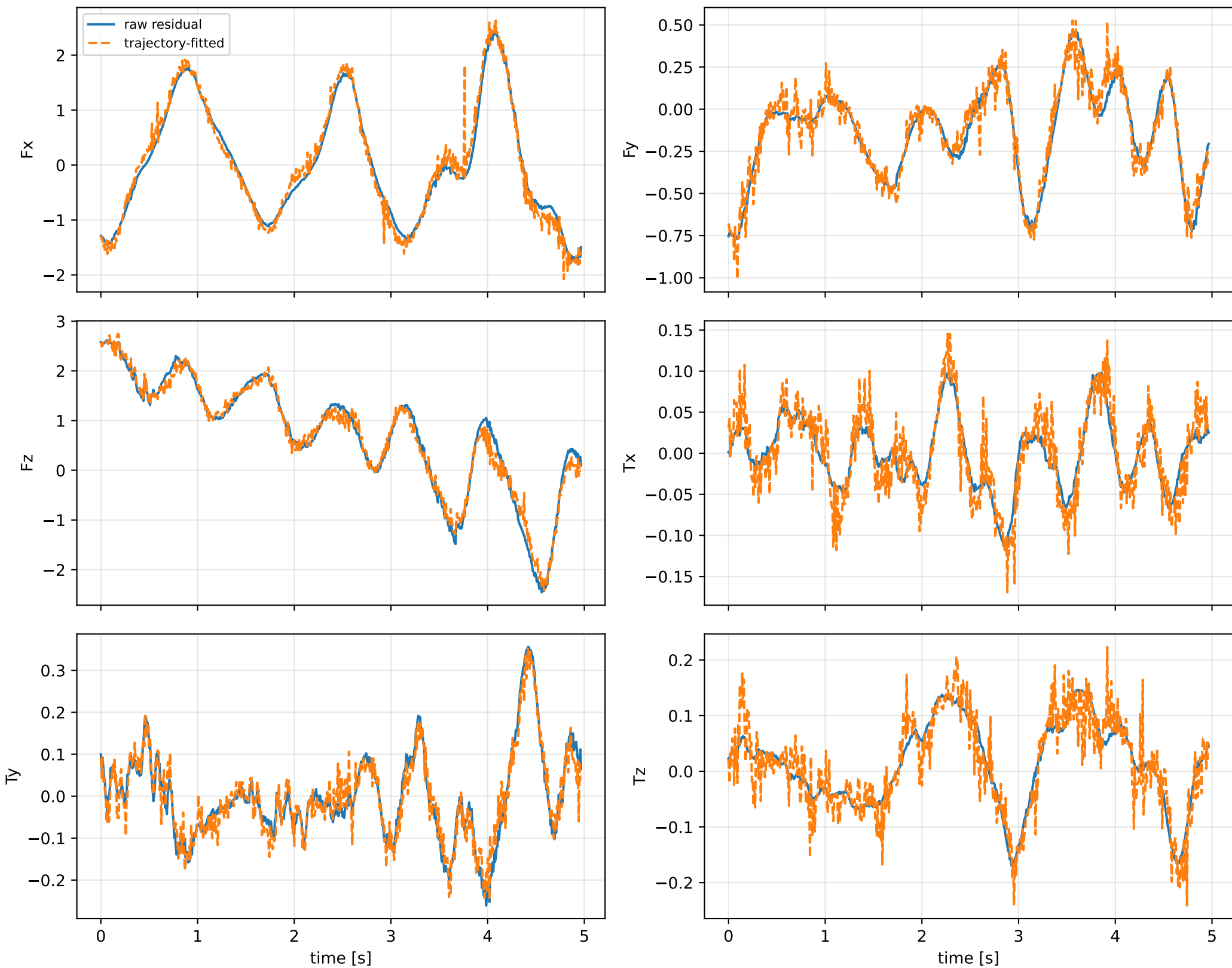
index 10: variance=0.05031, ratio=9.207
 $\log_{\text{mass}}=+0.000915$; second_moment_diag_1=-0.00792
second_moment_diag_2=+0.0256; second_moment_diag_3=-0.0195
second_moment_offdiag_12=-0.00374; second_moment_offdiag_13=-0.00746
second_moment_offdiag_23=-0.999; cog_x=+0.00143
cog_y=-0.00314; cog_z=-6.3e-05
log_force_eff_1=+0.0042; log_force_eff_2=+0.0208
log_force_eff_3=-0.00739; log_force_eff_4=-0.0168

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_all_nominal: optimization vs reference covariance



inertia_over_mass_all_nominal: wrench / closure (force max $7.88\text{e-}15$, torque max $2.08\text{e-}16$)



inertia_over_mass_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/inertia_over_mas

SHA256: 696c77257c9aa3a12bfed1ee5eb82a98ad78c606342ec66cb8ccb4cf17bb8a36

data objective: 953.410037283

prior objective: 0.00198703121377

total objective: 953.412024314

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_all_nominal (inertia_over_mass_m2)

components: xx, yy, zz, xy, xz, yz

target: [2.76408097e-02 2.76206510e-02 5.48529694e-02 -3.09533934e-07
8.08602633e-06 2.51604719e-08]

std: [1.e-06 1.e-06 1.e-06 1.e-06 1.e-06 1.e-06]

final: [2.76408155e-02 2.76207135e-02 5.48529713e-02 -3.14169354e-07
8.08664052e-06 2.82472819e-08]

error: [5.76171602e-09 6.24975253e-08 1.87926767e-09 -4.63541987e-09
6.14189785e-10 3.08681000e-09]

standardized residual: [0.00576172 0.06249753 0.00187927 -0.00463542 0.00061419 0.00308681]

factor objective: 0.00198703121377